

Several Bilinear Algorithms for Matrix Multiplication Problems <3, P, Q>

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Abstract — We propose the exact bilinear algorithms for matrix multiplication <3, 4, 6; 56>, <3, 5, 5; 58>, <3, 4, 7; 66>, <3, 4, 8; 75>, and <3, 5, 7; 82>.

Introduction

The problem of designing a bilinear algorithm useful for fast matrix multiplication was formulated by Strassen [1] half a century ago. It can be reduced to solving a nonlinear system of equations of Brent for the coefficients of the algorithm [2]. In [3], the least squares method was proposed for solving this system with a large number of zero roots and solutions were found for the problems <N, P, Q>, $N \cdot P \cdot Q < 64$. These solutions are presented in [3] and [4].

Our method did not allow us to obtain the known <4, 4, 4; 49> by direct calculation. The problem of local extrema, which increases with the size of the system, also hindered our progress. It was assumed that with an increase in size of problems, it is impossible to obtain short exact solutions due to local extrema and the convergence of the properties of solutions with polynomial orders 1 and 0. Subsequently, an effective method was found for getting out of local extrema and it turned out that for problems of the form <3, P, Q> it is possible to obtain short exact solutions even for large sizes.

Below we present algorithms that are shorter than the sums of already known algorithms. The results are presented in non-optimized VB code and with traditional indexation for the coefficients of result matrix. In another form, algorithms are presented or will be presented in [5].

In practical realization of these algorithms the calculation of similar linear forms can and should be optimized.

Algorithms

<3, 4, 6; 56>

$D1 = (-A(2,1) + A(2,2) + A(2,3) - A(2,4) + A(3,1) + A(3,2) + A(3,3) + A(3,4)) * (-B(1,3) + B(1,6) + B(2,1) - 0.5 * B(3,1) + 0.5 * B(3,6) + 0.5 * B(4,1) - 0.5 * B(4,6))$
 $D2 = (-A(1,2) + A(1,4) + A(3,3) + A(3,4)) * (B(1,3) - B(1,6) - B(2,1) + B(3,1) - B(3,3) - B(3,5) + B(3,6) + B(4,5))$
 $D3 = (A(2,1) + A(2,2) - A(3,1)) * (-B(1,6) + B(2,2) + B(2,4) + B(2,6) - B(3,2) - B(3,4) - B(3,6) + B(4,6))$
 $D4 = (-A(2,1) - A(2,2) + A(3,1) + A(3,2) + A(3,3) + A(3,4)) * (-B(2,1) + B(3,1) + B(4,6))$
 $D5 = (-A(2,1) - A(2,2) + A(3,1) + A(3,2)) * (B(2,1) + B(2,2) + B(2,4) + B(2,6) - B(3,1) - B(3,2) - B(3,4) - B(3,6))$
 $D6 = (A(2,1) + A(2,2) + A(2,3) + A(2,4) - A(3,1) - A(3,2) - A(3,3) - A(3,4)) * (0.5 * B(3,1) + 0.5 * B(3,6) + 0.5 * B(4,1) + 0.5 * B(4,6))$
 $D7 = (-A(2,1) + A(2,2) + A(2,3) - A(2,4) + A(3,1)) * (B(1,1) - B(2,1) + B(3,1) + B(3,2) + B(3,4) - B(4,1))$
 $D8 = (-A(1,2) + A(1,4) - A(3,2)) * (B(1,3) - B(1,6) - B(2,3) + B(2,6) + B(3,3) - B(3,6) + B(4,2) - B(4,3) + B(4,6))$
 $D9 = (A(2,1) - A(2,2) - A(2,3) + A(2,4) - A(3,1) + A(3,2) + A(3,3) - A(3,4)) * (0.5 * B(3,1) + B(3,2) + B(3,4) + 0.5 * B(3,6) - 0.5 * B(4,1) - 0.5 * B(4,6))$
 $D10 = (A(1,4) - A(2,2) - A(2,3) + A(2,4)) * (-B(1,3) + B(1,6) + B(2,1) - B(3,3) + B(3,6) + B(4,1))$
 $D11 = (A(1,4) + A(2,2)) * (B(2,1) - B(2,3) + B(2,6) - B(3,1) + B(3,3) - B(3,6) + B(4,6))$
 $D12 = (A(3,1) + A(3,2) + A(3,3) + A(3,4)) * (B(1,3) - B(1,6))$
 $D13 = (A(2,2) + A(2,3) - A(2,4)) * (-B(1,1) + B(2,1) - B(3,1) + B(4,1))$
 $D14 = (A(1,1) + A(1,2) - A(3,3) - A(3,4)) * (B(1,3) - B(1,6) + B(4,5))$
 $D15 = (A(2,1) - A(2,2) - A(2,3) + A(2,4)) * (B(1,1) - B(1,3) + B(1,6))$
 $D16 = (A(1,2) + A(2,2)) * (-B(2,1) + B(2,3) - B(2,4) + B(2,5) + B(3,1) - B(3,3) - B(3,5) + B(3,6))$

$$\begin{aligned}
D17 &= (-A(2,1) + A(3,1)) * (-B(1,1) - B(1,2) - B(1,4) - B(1,6) + B(2,1) + B(2,2) + B(2,4) + B(2,6) - B(3,1) - B(3,2) - B(3,4) - B(3,6) + B(4,1) + B(4,2) + B(4,4) + B(4,6)) \\
D18 &= (A(1,1) + A(2,2) + A(2,3) - A(2,4)) * (B(1,1) - B(1,3) + B(1,6) + B(3,5)) \\
D19 &= (A(2,1) + A(2,4) - A(3,1)) * (-B(1,4) + B(3,2) + B(3,4) + B(4,2) + B(4,4)) \\
D20 &= (A(1,2) + A(3,2)) * (-B(1,2) + B(1,3) + B(1,5) - B(1,6) + B(2,2) - B(2,3) - B(2,5) + B(2,6) + B(3,3) + B(3,5) - B(3,6) + B(4,2) - B(4,3) - B(4,5) + B(4,6)) \\
D21 &= (A(1,1) + A(1,2) + A(3,2)) * (B(1,2) - B(1,5) + B(2,5) - B(3,5) + B(4,5)) \\
D22 &= (A(1,2) - A(1,4) + A(3,2) - A(3,4)) * (B(3,3) + B(3,5) - B(3,6) + B(4,2) - B(4,3) - B(4,5) + B(4,6)) \\
D23 &= (-A(1,2) + A(1,4)) * (-B(2,1) + B(2,3) - B(2,6) + B(3,1) - B(3,3) + B(3,6)) \\
D24 &= (A(1,3) + A(1,4) + A(3,3) + A(3,4)) * (B(3,3) + B(3,5) - B(3,6)) \\
D25 &= (A(1,4)) * (-B(1,3) + B(1,6) + B(2,3) - B(2,6) - B(3,3) + B(3,6) + B(4,3) - B(4,6)) \\
D26 &= (A(1,2) + A(1,3) - A(1,4) + A(2,2) + A(2,3) - A(2,4)) * (B(3,1) - B(3,3) - B(3,5) + B(3,6)) \\
D27 &= (A(2,2)) * (B(1,6) - B(2,6) + B(3,6) - B(4,6)) \\
D28 &= (A(1,1) - A(2,2)) * (B(1,6) + B(2,5) - B(3,5)) \\
D29 &= (A(1,3) - A(3,2)) * (-B(2,4) + B(3,2) + B(3,4)) \\
D30 &= (-A(1,3) + A(2,2)) * (B(2,4) - B(3,6)) \\
D31 &= (A(1,1) + A(1,4)) * (B(3,5) + B(4,5)) \\
D32 &= (A(1,1) + A(1,2) + A(3,1) + A(3,2)) * (-B(1,2) + B(1,3) + B(1,5) - B(1,6)) \\
D33 &= (A(3,3) + A(3,4)) * (B(2,1) - B(3,1)) \\
D34 &= (A(3,2)) * (-B(2,2) - B(2,4) + B(3,2) + B(3,4)) \\
D35 &= (-A(2,1) - A(2,4) + A(3,1) + A(3,4)) * (B(3,2) + B(3,4) + B(4,2) + B(4,4)) \\
D36 &= (A(1,1) + A(1,2)) * (-B(2,5) + B(3,5)) \\
D37 &= (A(3,2) + A(3,3) - A(3,4)) * (B(3,2) + B(3,4)) \\
D38 &= (A(1,3)) * (B(2,4) - B(3,4)) \\
D39 &= (-A(1,1) + A(2,4)) * (-B(1,4) + B(3,5) + B(4,5)) \\
D40 &= (A(3,2) + A(3,3)) * (B(3,6) - B(4,6)) \\
D41 &= (A(1,1)) * (B(1,5) - B(2,5) + B(3,5) - B(4,5)) \\
D42 &= (A(2,1) + A(2,2)) * (B(1,6)) \\
D43 &= (A(1,4) + A(2,4)) * (-B(3,1) + B(3,3) - B(3,6) - B(4,1) + B(4,3)) \\
D44 &= (A(1,1) + A(2,1)) * (-B(1,1) + B(1,3) - B(1,4) + B(1,5)) \\
D45 &= (A(3,2) + A(3,3)) * (-B(3,5) + B(4,5)) \\
D46 &= (A(2,2) + A(2,3)) * (B(3,6)) \\
D47 &= (A(1,3) + A(2,3)) * (-B(3,4) + B(3,6)) \\
D48 &= (A(1,1) - A(1,2) - A(1,3) + A(1,4)) * (B(3,5)) \\
D49 &= (A(1,4) + A(2,4)) * (B(3,5) - B(4,4) + B(4,5)) \\
D50 &= (A(2,4)) * (B(1,4) - B(4,4)) \\
D51 &= (A(2,1) + A(2,4)) * (B(1,4)) \\
D52 &= (A(1,2) + A(1,3)) * (B(2,4)) \\
D53 &= (A(2,2) - A(2,4)) * (B(4,6)) \\
D54 &= (A(1,3) + A(3,3)) * (B(3,2)) \\
D55 &= (A(3,4)) * (B(4,2)) \\
D56 &= (A(3,1)) * (B(1,2))
\end{aligned}$$

$$\begin{aligned}
C(1,1) &= D2 + D10 + D13 + D14 + D18 + D24 + D26 - D31 + D33 \\
C(1,2) &= D8 + D20 + D21 + D25 + D29 + D34 + D38 + D41 \\
C(1,3) &= D2 + D11 + D14 + D16 + D24 + D25 + D27 + D28 + D30 - D31 + D33 + D36 + D48 + D52 \\
C(1,4) &= D31 - D38 + D39 - D49 + D50 + D52 \\
C(1,5) &= D31 - D36 + D41 - D48 \\
C(1,6) &= D11 + D16 + D23 + D27 + D28 + D30 + D36 + D52 \\
C(2,1) &= D1 + D4 + D6 + D12 - D13 + D15 - D46 + D53 \\
C(2,2) &= D3 + D7 + D13 + D17 + D19 + D27 - D30 - D38 - D46 + D47 + D50 + D56 \\
C(2,3) &= D1 + D4 + D6 + D10 - D11 + D12 - D25 - D27 + D42 + D43 + D53 \\
C(2,4) &= D30 + D38 + D46 - D47 - D50 + D51 \\
C(2,5) &= D15 + D18 - D28 + D39 - D41 - D42 + D44 + D51 \\
C(2,6) &= -D27 + D42 + D46 - D53 \\
C(3,1) &= D1 + D7 + D9 + D12 + D15 - D33 - D37 - D40 \\
C(3,2) &= -D29 - D34 - D38 + D54 + D55 + D56 \\
C(3,3) &= -D2 + D3 + D4 + D5 + D8 + D12 + D22 + D23 + D34 + D40 + D42 + D45 + D55 \\
C(3,4) &= D19 + D29 + D35 + D37 + D38 + D51 - D54 - D55
\end{aligned}$$

$$C(3,5) = -D_{12} - D_{14} + D_{21} + D_{32} + D_{36} - D_{45} + D_{56}$$

$$C(3,6) = D_3 + D_4 + D_5 + D_{33} + D_{34} + D_{40} + D_{42}$$

<3, 5, 5; 58> [6]

$$\begin{aligned} D_1 &= (A(1,3) + A(1,4) + A(3,4)) * (B(3,3) + B(3,5) + B(4,2)) \\ D_2 &= (A(1,2) + A(2,4)) * (-B(1,3) - B(2,2) - B(2,3) + B(2,4) - B(3,3) + B(4,3)) \\ D_3 &= (-A(1,3) - A(1,4) + A(2,1) + A(2,3) + A(2,4)) * (B(1,2) + B(3,5)) \\ D_4 &= (A(1,1) + A(1,4)) * (B(3,5) - B(4,5)) \\ D_5 &= (A(1,1) + A(1,4) + A(3,4)) * (B(1,3) - B(3,5) + B(4,5)) \\ D_6 &= (A(1,1) + A(1,4) - A(2,4)) * (B(1,4) - B(3,5) + B(4,5)) \\ D_7 &= (A(1,2) - A(2,2) + A(2,3)) * (B(2,2) + B(3,1)) \\ D_8 &= (A(2,5) - A(3,3) - A(3,5)) * (B(1,2) - B(3,2) + B(3,4) + B(5,2)) \\ D_9 &= (-A(1,4) + A(1,5) + A(2,4)) * (-B(4,1) - B(5,1) + B(5,4) + B(5,5)) \\ D_{10} &= (A(1,3) + A(1,4)) * (B(3,5) + B(4,2)) \\ D_{11} &= (A(1,5) + A(3,3) + A(3,5)) * (-B(3,1) + B(5,2)) \\ D_{12} &= (-A(2,2) + A(2,3) + A(3,2)) * (B(2,2) - B(3,3)) \\ D_{13} &= (A(1,3) + A(1,4) + A(2,2) - A(2,3) - A(3,2) + A(3,3) + A(3,4)) * B(3,3) \\ D_{14} &= (-A(1,2) + A(1,3) + A(1,5) + A(2,2) - A(2,3) + A(3,3) + A(3,5)) * B(3,1) \\ D_{15} &= (A(1,3) + A(1,4) - A(2,3) - A(2,4) - A(2,5) + A(3,3) + A(3,5)) * (B(1,2) - B(3,2) + B(3,4)) \\ D_{16} &= (A(2,2) - A(2,3)) * B(2,2) \\ D_{17} &= (A(2,1) - A(2,2) + A(3,2)) * (B(1,3) - B(2,5)) \\ D_{18} &= (A(1,2) + A(2,1) - A(2,2)) * (B(1,1) + B(2,5)) \\ D_{19} &= (-A(2,5) + A(3,1) + A(3,5)) * (B(1,4) + B(5,5)) \\ D_{20} &= (-A(1,4) + A(2,4)) * (B(1,4) - B(3,2) + B(3,4) + B(4,1) + B(4,2) - B(4,4) + B(5,1) - B(5,4) - B(5,5)) \\ D_{21} &= (A(1,3) + A(1,4) - A(2,4)) * (-B(3,2) + B(3,4) + B(4,2)) \\ D_{22} &= (A(1,5) + A(3,1) + A(3,5)) * (-B(1,1) + B(5,5)) \\ D_{23} &= A(2,4) * (B(1,3) + B(2,2) + B(2,3) - B(2,4) + B(3,3) - B(3,5) - B(4,3) + B(4,4) + B(4,5)) \\ D_{24} &= (-A(1,2) + A(2,2) + A(2,5)) * (-B(2,2) + B(2,4)) \\ D_{25} &= A(1,2) * (B(1,3) + B(2,3) + B(3,3) - B(4,3)) \\ D_{26} &= (A(2,2) + A(2,4) - A(3,2)) * (B(1,3) + B(2,2) + B(2,3) - B(2,4) + B(3,3)) \\ D_{27} &= (A(1,1) + A(1,4) - A(2,1) + A(2,2) + A(3,1) - A(3,2) + A(3,4)) * (B(1,3)) \\ D_{28} &= -(A(3,1) + A(3,5)) * B(5,5) \\ D_{29} &= (-A(2,1) + A(2,2)) * B(2,5) \\ D_{30} &= (A(2,5) - A(3,5)) * (B(1,2) + B(1,4) - B(3,2) + B(3,4) + B(5,2) + B(5,3) - B(5,4)) \\ D_{31} &= (A(1,1) - A(1,2) + A(1,5) - A(2,1) + A(2,2) + A(3,1) + A(3,5)) * B(1,1) \\ D_{32} &= (A(1,1) + A(1,4) - A(2,1) - A(2,4) - A(2,5) + A(3,1) + A(3,5)) * B(1,4) \\ D_{33} &= (A(2,5) + A(3,2)) * (B(2,2) - B(2,4) + B(5,3)) \\ D_{34} &= (-A(1,4) + A(1,5) + A(2,4) - A(2,5)) * (-B(5,1) + B(5,4) + B(5,5)) \\ D_{35} &= A(1,5) * (B(4,1) + B(5,1) - B(5,2) - B(5,5)) \\ D_{36} &= (A(3,3) + A(3,5)) * B(5,2) \\ D_{37} &= (A(1,1) + A(1,3) + A(1,4) - A(2,1) - A(2,3) - A(2,4)) * (B(1,2)) \\ D_{38} &= A(2,5) * (B(2,2) - B(2,4) + B(5,4) + B(5,5)) \\ D_{39} &= (A(2,1) + A(2,3) + A(2,4)) * B(3,5) \\ D_{40} &= (A(1,2) + A(1,4) + A(3,4)) * (B(1,3) + B(3,3) + B(4,1) - B(4,3)) \\ D_{41} &= (-A(1,2) + A(2,2)) * (B(1,1) + B(2,1) + B(2,2) - B(2,4) + B(3,1)) \\ D_{42} &= A(3,3) * (-B(1,2) + B(3,2) + B(3,3) + B(3,5) - B(5,2)) \\ D_{43} &= (A(3,2) - A(3,4) + A(3,5)) * B(5,3) \\ D_{44} &= (A(1,2) + A(1,4) - A(1,5)) * B(4,1) \\ D_{45} &= (A(1,5) + A(3,5)) * (B(1,1) + B(3,1) - B(5,1) + B(5,3)) \\ D_{46} &= (A(1,5) + A(3,4)) * (B(4,1) + B(5,3)) \\ D_{47} &= A(3,2) * (B(2,2) + B(2,3) + B(2,5) - B(5,3)) \\ D_{48} &= -A(1,2) * (-B(2,1) + B(2,2) + B(2,5) + B(4,1)) \\ D_{49} &= -A(3,4) * (B(4,2) + B(4,3) + B(4,5) + B(5,3)) \\ D_{50} &= A(1,3) * (-B(3,1) + B(3,2) - B(4,2)) \\ D_{51} &= -(A(3,3) + A(3,4)) * B(3,5) \\ D_{52} &= A(1,1) * (-B(1,1) + B(1,2) + B(1,5) + B(3,5) - B(4,5)) \\ D_{53} &= A(2,1) * (B(1,4) + B(1,5) + B(2,5) - B(3,5)) \\ D_{54} &= A(2,3) * (B(2,2) + B(3,4)) \end{aligned}$$

$$\begin{aligned} D55 &= A(3,4) * (-B(4,2) + B(4,4) + B(5,3)) \\ D56 &= A(3,1) * (B(1,3) + B(1,5) - B(5,5)) \\ D57 &= A(3,2) * (B(2,1) - B(5,3)) \\ D58 &= (A(3,1) + A(3,3)) * B(1,2) \end{aligned}$$

$$\begin{aligned} C(1,1) &= D7 + D11 + D14 + D16 + D18 + D22 + D28 + D29 + D31 + D35 - D36 + D44 + D48 \\ C(1,2) &= D3 + D7 + D10 + D11 + D14 + D16 - D36 + D37 - D39 + D50 \\ C(1,3) &= D1 + D4 + D5 - D10 + D25 - D40 + D44 + D46 + D49 \\ C(1,4) &= D2 + D4 + D6 + D7 + D9 + D11 + D14 + D16 + D20 + D21 + D23 + D25 + D35 - D36 + D50 \\ C(1,5) &= -D3 - D4 + D18 + D22 + D28 + D29 + D31 - D37 + D39 + D52 \\ C(2,1) &= D7 - D9 + D16 + D18 + D24 + D29 + D34 + D38 + D41 + D44 + D48 \\ C(2,2) &= D3 + D8 + D10 + D15 + D16 - D21 + D36 - D39 + D54 \\ C(2,3) &= D2 - D12 - D16 + D17 + D24 + D25 + D26 - D29 + D33 + D47 \\ C(2,4) &= D2 + D4 + D6 + D16 + D19 + D23 + D24 + D25 + D28 - D32 + D38 + D54 \\ C(2,5) &= -D4 - D6 - D19 - D28 + D29 + D32 + D39 + D53 \\ C(3,1) &= -D11 - D22 - D28 - D35 + D36 + D43 - D45 + D46 + D57 \\ C(3,2) &= D1 - D10 + D12 - D13 + D16 + D36 + D42 + D51 + D58 \\ C(3,3) &= -D1 - D4 - D5 + D10 - D12 + D13 - D16 + D17 + D27 - D29 + D43 + D47 - D49 \\ C(3,4) &= D1 - D8 - D10 + D12 - D13 + D16 + D19 + D28 + D30 - D33 + D38 + D42 + D43 + D51 + D55 \\ C(3,5) &= D4 + D5 - D17 - D27 - D28 + D29 - D51 + D56 \end{aligned}$$

<3, 4, 7; 66>

$$\begin{aligned} D1 &= (-A(2,4) + A(3,3) + A(3,4)) * (-B(1,2) + B(1,7) + B(3,7) - B(4,4)) \\ D2 &= (-A(1,3) - A(2,4) + A(3,3) + A(3,4)) * (B(1,2) - B(1,7) - B(3,7) - B(4,1) + B(4,5) + B(4,7)) \\ D3 &= (-A(2,1) + A(3,1) + A(3,3)) * (-B(1,2) + B(1,4) + B(3,6) - B(4,6)) \\ D4 &= (-A(1,1) + A(3,1) + A(3,3)) * (-B(1,2) + B(1,4) + B(3,5) - B(4,5)) \\ D5 &= (-A(1,1) + A(3,3) + A(3,4)) * (B(1,2) + B(4,5)) \\ D6 &= (A(3,1) + A(3,3)) * (B(1,2) - B(1,4)) \\ D7 &= (A(3,3) + A(3,4)) * (B(1,2) + B(4,4)) \\ D8 &= (A(1,3) + A(2,2) - A(2,3) - A(3,2)) * (-B(2,2) + B(2,3) + B(2,5) + B(3,3)) \\ D9 &= (A(2,3) + A(2,4) - A(3,3) - A(3,4)) * (-B(1,2) + B(1,7) + B(3,7) + B(4,6)) \\ D10 &= (-A(2,1) + A(3,3) + A(3,4)) * (-B(1,2) + B(4,6)) \\ D11 &= (-A(1,1) + A(1,3) + A(2,4)) * (B(1,2) - B(1,7) + B(4,5)) \\ D12 &= (A(1,2) - A(2,2) + A(2,3)) * (B(2,4) + B(3,1)) \\ D13 &= (-A(2,2) + A(2,3) + A(3,2)) * (-B(2,4) + B(3,3)) \\ D14 &= (A(1,3) + A(1,4) - A(3,3) - A(3,4)) * (B(3,2) - B(4,5)) \\ D15 &= (-A(1,4) + A(3,3) + A(3,4)) * (B(3,2) - B(4,4)) \\ D16 &= (-A(2,1) - A(2,3) + A(3,1) + A(3,3)) * (-B(3,6) + B(4,6)) \\ D17 &= (A(1,1) + A(1,3) - A(3,1) - A(3,3)) * (B(3,5) - B(4,5)) \\ D18 &= (A(2,2) - A(2,3)) * (B(2,4)) \\ D19 &= (-A(2,2) + A(3,2)) * (B(2,3) - B(2,6) + B(2,7) + B(3,3)) \\ D20 &= (A(1,4) + A(2,3) - A(3,3) - A(3,4)) * (B(3,2) + B(4,6)) \\ D21 &= (A(2,4) - A(3,2)) * (-B(2,7) + B(4,3)) \\ D22 &= (A(1,1)) * (-B(1,2) + B(1,4) + B(1,5) + B(2,5) + B(4,5)) \\ D23 &= (A(2,1)) * (-B(1,2) + B(1,4) + B(1,6) + B(2,6) + B(4,6)) \\ D24 &= (A(2,4)) * (-B(1,2) + B(1,7) + B(2,7) - B(4,4) + B(4,7)) \\ D25 &= (-A(1,3) + A(2,3)) * (B(1,2) - B(1,7) - B(2,2) + B(2,3) + B(2,5) + B(3,3) + B(3,5)) \\ D26 &= (-A(2,4) + A(3,4)) * (-B(3,7) + B(4,3) + B(4,7)) \\ D27 &= (-A(1,2) + A(1,3) + A(2,2) - A(2,3)) * (B(3,1)) \\ D28 &= (A(1,2) - A(1,3) - A(2,2) + A(2,3)) * (-B(2,2) + B(2,3) + B(2,5)) \\ D29 &= (-A(2,1) + A(2,3) + A(2,4)) * (-B(4,6)) \\ D30 &= (-A(1,2) + A(2,4)) * (B(2,7) + B(4,1)) \\ D31 &= (-A(1,1) + A(1,3) + A(1,4)) * (B(4,5)) \\ D32 &= (-A(2,1) + A(2,3) + A(2,4)) * (B(1,2) - B(1,7)) \\ D33 &= (A(1,1) - A(3,2)) * (B(1,3) - B(2,5)) \\ D34 &= (A(3,2)) * (B(1,3) + B(2,3) + B(2,4) + B(4,3)) \\ D35 &= (A(3,1) - A(3,3) - A(3,4)) * (B(1,2)) \\ D36 &= (A(1,2)) * (B(1,1) + B(2,1) - B(2,4) + B(4,1)) \end{aligned}$$

$D37 = (-A(1,4) + A(3,2)) * (B(2,2) + B(4,3))$
 $D38 = (-A(2,1) + A(3,2)) * (B(1,3) + B(2,6))$
 $D39 = (-A(2,1) + A(3,1)) * (B(1,2) - B(1,3) + B(1,6) - B(3,6) + B(4,6))$
 $D40 = (-A(1,1) + A(3,1)) * (-B(1,2) + B(1,3) + B(1,5) - B(3,5) + B(4,5))$
 $D41 = (A(1,4)) * (B(2,2) + B(4,2) - B(4,4))$
 $D42 = (A(2,2) - A(2,3) - A(3,2) + A(3,3)) * (B(3,3))$
 $D43 = (A(1,3) + A(1,4) - A(3,3) - A(3,4)) * (-B(4,1) + B(4,7))$
 $D44 = (A(2,2) - A(3,2)) * (B(2,2) + B(2,7))$
 $D45 = (-A(1,3) + A(3,3)) * (B(3,1) - B(3,7) - B(4,1) + B(4,7))$
 $D46 = (-A(2,1) + A(2,2)) * (B(2,6))$
 $D47 = (-A(1,2) + A(2,1)) * (B(1,1) + B(2,6))$
 $D48 = (A(1,2) - A(2,2)) * (-B(2,1) + B(2,6) + B(2,7) - B(3,1))$
 $D49 = (-A(3,2) + A(3,4)) * (B(4,3))$
 $D50 = (-A(2,2) + A(2,4)) * (B(2,7))$
 $D51 = (-A(1,1) + A(2,1)) * (-B(1,2) + B(1,3) + B(1,5) + B(1,7))$
 $D52 = (A(2,3)) * (B(1,2) - B(1,7) + B(2,4) + B(3,4) + B(3,6) - B(3,7) - B(4,6))$
 $D53 = (A(1,1) - A(1,2)) * (B(1,1))$
 $D54 = (A(1,4) - A(3,4)) * (B(3,2) - B(4,2) + B(4,3))$
 $D55 = (-A(1,4) + A(2,4)) * (B(4,2) - B(4,3) + B(4,6))$
 $D56 = (A(3,1) - A(3,3) - A(3,4)) * (B(1,7))$
 $D57 = (A(1,1) - A(2,1)) * (-B(1,1) + B(1,2) + B(1,6))$
 $D58 = (A(3,1) - A(3,2)) * (B(1,3))$
 $D59 = (A(1,3)) * (-B(3,1) - B(3,2) + B(3,4) + B(3,5) - B(4,5))$
 $D60 = (A(3,3)) * (B(1,2) - B(1,4) + B(3,3) + B(3,4) - B(4,4))$
 $D61 = (A(1,1) - A(3,1)) * (-B(1,1) + B(1,7))$
 $D62 = (-A(1,1) + A(1,2)) * (B(2,5))$
 $D63 = (A(1,2) - A(1,4)) * (B(2,2))$
 $D64 = (-A(1,2) + A(1,4)) * (B(4,1))$
 $D65 = (A(1,2) - A(3,2)) * (-B(2,1) + B(2,7))$
 $D66 = (A(1,3) - A(2,3)) * (B(3,2) + B(3,6))$

$C(1,1) = D12 + D18 + D27 + D36 + D53 + D64$
 $C(1,2) = -D5 + D7 + D14 + D15 + D31 + D41 + D63$
 $C(1,3) = D8 + D13 - D18 + D28 + D33 + D34 - D37 - D62 + D63$
 $C(1,4) = -D4 - D5 - D6 + D7 + D12 + D14 + D15 - D17 + D18 + D27 + D31 + D59$
 $C(1,5) = D4 + D6 + D17 + D22 + D31 + D62$
 $C(1,6) = D3 + D5 + D6 + D10 - D14 + D16 + D20 + D23 - D31 - D47 + D53 + D57 + D66$
 $C(1,7) = D1 + D2 - D5 + D7 + D11 + D24 - D30 + D43 + D64$
 $C(2,1) = D12 + D18 + D30 + D36 + D46 + D47 + D48 - D50$
 $C(2,2) = D7 + D10 + D15 + D20 + D21 + D29 + D37 + D41 + D44 + D50 + D55$
 $C(2,3) = D13 - D18 - D19 + D21 + D34 - D38 + D46 + D50$
 $C(2,4) = D1 - D3 - D6 + D7 + D9 + D10 - D16 + D18 + D29 + D52$
 $C(2,5) = D4 + D6 + D8 + D11 + D17 + D19 + D22 + D25 - D32 + D33 + D38 + D44 - D46 + D51$
 $C(2,6) = D3 + D6 + D16 + D23 - D29 + D46$
 $C(2,7) = D1 + D7 + D9 + D10 + D24 + D29 + D32 - D50$
 $C(3,1) = -D2 + D5 - D11 + D12 + D18 + D21 + D26 + D27 + D30 + D36 + D45 - D49 + D53 + D56 + D61 + D65$
 $C(3,2) = D7 + D15 + D35 + D37 + D41 + D49 + D54$
 $C(3,3) = D13 - D18 + D34 + D42 + D49 + D58$
 $C(3,4) = -D6 + D7 - D13 + D18 + D35 - D42 + D60$
 $C(3,5) = D4 + D5 + D6 + D22 + D33 + D35 + D40 - D58$
 $C(3,6) = D3 + D6 + D10 + D23 - D35 + D38 + D39 + D58$
 $C(3,7) = D1 + D7 + D21 + D24 + D26 - D49 + D56$

<3, 4, 8; 75>

$D1 = (A(1,1) + A(1,3) - A(2,1) - A(2,3) + A(3,1) + A(3,3)) * (B(1,1) - B(1,2) - B(1,3) + B(1,4) + B(1,7) - B(1,8) + B(3,1) + B(3,4) + B(3,7))$
 $D2 = (-A(1,1) + A(1,3) + A(2,1) - A(2,3) - A(3,1) + A(3,3)) * (-0.5 * B(1,1) + 0.5 * B(1,2) + 0.5 * B(1,3) - 0.5 * B(1,4) - 0.5 * B(1,7) + 0.5 * B(1,8) + 0.5 * B(3,1) + 0.5 * B(3,4) + 0.5 * B(3,7))$

$D3 = (A(1,2) + A(2,4) - A(3,2) - A(3,4)) * (-B(2,2) + B(2,7) - B(4,7) + B(4,8))$
 $D4 = (A(1,3) - A(2,3) + A(3,3)) * (B(1,1) - B(1,2) - B(1,3) + B(1,4) + B(1,7) - B(1,8) + B(3,1))$
 $D5 = (A(1,2) + A(2,4) - A(3,4)) * (0.5 * B(2,2) - 0.5 * B(2,7) - B(4,3))$
 $D6 = (-A(1,3) + A(2,3) + A(3,1) - A(3,3)) * (B(1,1) - B(1,2) - B(1,3) + B(1,4) + B(1,7) - B(1,8) + B(3,5))$
 $D7 = (A(1,2) + A(1,4) - A(3,2) - A(3,4)) * (0.5 * B(4,7) - 0.5 * B(4,8))$
 $D8 = (A(1,1) - A(2,1) + A(3,1)) * (B(1,1) - B(1,2) + B(1,7) - B(1,8) + B(3,1) + B(3,4) + B(3,7))$
 $D9 = (A(1,2) - A(2,2) + A(3,2)) * (B(2,2) + B(2,7))$
 $D10 = (A(1,1) - A(2,1) + A(2,3) + A(3,1) - A(3,3)) * (-B(1,6) + B(3,1) + B(3,4) + B(3,7))$
 $D11 = (-A(1,4) + A(2,4)) * (B(4,3) + 0.5 * B(4,7) - 0.5 * B(4,8))$
 $D12 = (A(1,4) - A(2,1) + A(3,1)) * (B(1,7) + B(4,4) + 0.5 * B(4,7) + 0.5 * B(4,8))$
 $D13 = (-A(2,1) - A(2,2) + A(3,2)) * (B(1,7) - B(1,8) + B(2,5))$
 $D14 = (-A(1,3) + A(2,3) + A(3,2)) * (B(2,1) + B(2,2) - B(2,8) - B(3,7) + B(3,8))$
 $D15 = (A(1,3) - A(2,3)) * (B(2,1) + B(2,2) - B(2,8) - B(3,1) + B(3,2) + B(3,5) - B(3,7) + B(3,8) - B(4,1) + B(4,2) - B(4,3))$
 $D16 = (A(1,4) + A(2,3) - A(2,4)) * (B(3,7) - B(4,6))$
 $D17 = (-A(1,2) + A(2,2)) * (B(2,2) + B(2,7))$
 $D18 = (A(1,3) - A(2,3) + A(3,4)) * (-B(3,2) + B(4,1) - B(4,2) + B(4,3))$
 $D19 = (A(1,2) - A(3,2)) * (B(2,7) - 0.5 * B(4,7) + 0.5 * B(4,8))$
 $D20 = (-A(1,2) + A(2,2) + A(3,3)) * (B(2,2) - B(2,3) + B(2,4) + B(3,3))$
 $D21 = (A(1,2) - A(2,2)) * (B(2,2) - B(2,3) + B(2,5) + B(3,3))$
 $D22 = (A(2,2) + A(2,4) - A(3,2) - A(3,4)) * (B(2,2) - B(2,7))$
 $D23 = (-A(1,4) + A(2,4)) * (B(3,7) + B(4,5) + 0.5 * B(4,7) + 0.5 * B(4,8))$
 $D24 = (A(2,1) - A(2,4) + A(3,4)) * (B(1,2) + B(4,5))$
 $D25 = (A(2,4) - A(3,4)) * (B(1,2) - B(2,2) + B(2,7) + B(4,2) - B(4,6) - 0.5 * B(4,7) + 0.5 * B(4,8))$
 $D26 = (A(1,4)) * (B(1,7) + B(4,6) + 0.5 * B(4,7) + 0.5 * B(4,8))$
 $D27 = (A(2,1) + A(2,2) - A(3,1) - A(3,2)) * (B(1,7) - B(1,8))$
 $D28 = (-A(2,1) + A(2,3) + A(3,1) - A(3,3)) * (B(1,6))$
 $D29 = (A(2,1) - A(3,1)) * (B(1,4) + B(1,7) - B(2,4) + B(4,4) + 0.5 * B(4,7) + 0.5 * B(4,8))$
 $D30 = (A(1,2) + A(1,4)) * (B(4,3))$
 $D31 = (A(2,3) + A(3,1) - A(3,3)) * (-B(1,6) + B(3,5))$
 $D32 = (A(2,1) - A(2,4) - A(3,1) + A(3,4)) * (B(4,4) + 0.5 * B(4,7) + 0.5 * B(4,8))$
 $D33 = (A(1,2) + A(2,1) - A(3,1)) * (-B(1,3) + B(2,4))$
 $D34 = (-A(2,3) + A(2,4)) * (B(4,6))$
 $D35 = (-A(2,1) + A(2,4)) * (B(4,5))$
 $D36 = (A(2,1) + A(2,2)) * (B(2,5))$
 $D37 = (A(1,2) + A(2,1)) * (B(1,3) - B(2,5))$
 $D38 = (-A(3,2)) * (B(2,2) + B(2,5) - B(2,8) - B(3,7) + B(3,8))$
 $D39 = (A(1,4) - A(2,1)) * (-B(1,7) + B(4,5))$
 $D40 = (A(2,2) - A(3,2)) * (B(1,7) - B(1,8) - B(2,6) + B(2,7) + B(2,8))$
 $D41 = (A(2,3) - A(3,3)) * (B(3,4) + B(3,7))$
 $D42 = (A(1,4) - A(2,4) + A(3,3)) * (B(3,7) - B(4,4))$
 $D43 = (-A(1,2) + A(2,2) + A(2,3)) * (-B(2,6) + B(3,3))$
 $D44 = (A(1,1) + A(2,2) - A(3,2)) * (B(1,7) - B(1,8) + B(2,1))$
 $D45 = (A(2,3) + A(3,2)) * (B(2,6) + B(3,7) - B(3,8))$
 $D46 = (A(3,4)) * (B(4,3) + 0.5 * B(4,7) - 0.5 * B(4,8))$
 $D47 = (A(3,4)) * (B(3,2) + B(4,2) - B(4,3) - B(4,5))$
 $D48 = (-A(1,3) + A(1,4) + A(2,3) - A(2,4)) * (B(3,7))$
 $D49 = (-A(1,1) + A(2,4) - A(3,4)) * (B(1,2) + B(4,1))$
 $D50 = (A(3,3)) * (-B(2,2) + B(2,3) - B(2,4) - B(3,3) + B(3,4) + B(4,4))$
 $D51 = (A(1,3) - A(1,4) - A(2,3) + A(2,4)) * (B(4,1) - B(4,2) + B(4,3))$
 $D52 = (-A(2,1) + A(2,4) + A(3,1) - A(3,4)) * (B(1,2))$
 $D53 = (-A(1,2) - A(1,3) + A(2,2) + A(2,3)) * (B(2,1) + B(2,2) - B(2,8))$
 $D54 = (A(2,3) - A(3,4)) * (B(3,2) + B(4,6))$
 $D55 = (A(2,2) + A(2,3)) * (B(2,6))$
 $D56 = (-A(3,1) + A(3,3)) * (B(3,5))$
 $D57 = (A(1,1) - A(2,1)) * (-B(1,1) + B(1,2) - B(1,3) + B(1,5) + B(1,8))$
 $D58 = (A(3,1)) * (-B(1,3) + B(1,4))$
 $D59 = (-A(2,3) + A(3,3)) * (B(1,6) - B(3,2) + B(3,6) + B(3,7) - B(3,8))$
 $D60 = (-A(2,1) - A(2,2) + A(3,1) + A(3,2)) * (B(2,4))$
 $D61 = (A(3,2) + A(3,3)) * (-B(3,7) + B(3,8))$

$$\begin{aligned}
D62 &= (A(1,2) + A(1,3) - A(2,2) - A(2,3)) * (B(3,3)) \\
D63 &= (-A(1,2)) * (B(1,3) - B(2,3) + B(2,6) + B(4,3)) \\
D64 &= (A(1,1)) * (B(1,1) + B(1,6) - B(2,1) + B(4,1)) \\
D65 &= (A(2,1)) * (B(1,5) + B(1,6) - B(2,5) + B(4,5)) \\
D66 &= (A(2,3)) * (-B(2,6) + B(3,5) + B(3,6) + B(4,6)) \\
D67 &= (A(1,1) + A(1,2)) * (B(1,3)) \\
D68 &= (A(3,3) - A(3,4)) * (B(4,4)) \\
D69 &= (A(1,1) - A(1,4)) * (B(1,7)) \\
D70 &= (A(3,2) + A(3,3)) * (-B(2,2) + B(2,3) - B(2,4)) \\
D71 &= (A(3,1)) * (-B(1,2) + B(1,5) + B(1,7) - B(1,8) + B(3,5)) \\
D72 &= (A(1,1) + A(1,2)) * (B(2,1)) \\
D73 &= (-A(1,1) + A(1,4)) * (B(4,1)) \\
D74 &= (A(3,3) - A(3,4)) * (B(3,2)) \\
D75 &= (A(1,3)) * (B(3,1) - B(3,3) + B(3,6) + B(3,7))
\end{aligned}$$

$$\begin{aligned}
C(1,1) &= -0.5 * D1 + D2 + D4 + D10 + D28 - D41 + D64 + D72 + D73 \\
C(1,2) &= -0.5 * D3 + D5 + 0.5 * D9 + 0.5 * D17 - D18 + D19 + D25 + D30 + D34 - D49 + D51 + D54 + D73 \\
C(1,3) &= D30 + D43 + D55 + D62 + D63 + D67 \\
C(1,4) &= D1 - D4 - D8 + D12 - D16 - D26 + D29 + D33 + D34 + D41 + D48 + D67 \\
C(1,5) &= -0.5 * D1 + D2 - D6 + D8 + D28 + D31 - D37 + D39 + D57 + D58 + D65 + D67 - D69 \\
C(1,6) &= 0.5 * D1 - D2 - D4 - D10 - D16 - D28 + D34 + D41 + D43 + D48 + D55 + D62 + D75 \\
C(1,7) &= -0.5 * D3 - D5 + 0.5 * D9 - D11 + D16 + 0.5 * D17 + D19 + D26 - D30 - D34 + D46 - D48 + D69 \\
C(1,8) &= 0.5 * D3 + D5 + 0.5 * D9 + D11 - D14 + D16 - 0.5 * D17 - D19 + D26 + D30 - D34 + D40 - D44 - D45 - D46 - \\
&\quad D48 + D53 + D55 + D69 + D72 \\
C(2,1) &= D4 + D6 - D8 + D10 + D13 + D14 + D15 + D18 + D24 + D28 + D35 + D36 + D38 - D41 + D44 + D47 + D49 + \\
&\quad D56 - D58 + D64 \\
C(2,2) &= D5 + D7 + 0.5 * D9 + D11 + D17 + D19 + 0.5 * D22 + D24 + D25 + D30 + D34 + D35 + D47 + D54 \\
C(2,3) &= 0.5 * D3 + D7 + D11 + 0.5 * D17 + D21 + 0.5 * D22 + D30 + D36 + D37 + D43 + D55 + D63 \\
C(2,4) &= D12 - D16 + D20 + D21 - D26 + D29 + D33 + D34 + D36 + D37 + D41 + D42 + D50 + D58 \\
C(2,5) &= D28 + D31 + D35 + D36 + D56 + D65 \\
C(2,6) &= -D28 - D31 + D34 + D55 - D56 + D66 \\
C(2,7) &= -D3 - D5 - D7 + 0.5 * D9 - D11 + D16 + D17 + D19 - 0.5 * D22 + D23 + D26 - D30 - D34 - D35 + D39 + D46 \\
C(2,8) &= D3 + D5 + D7 + 0.5 * D9 + D11 + D13 + D16 - D19 + 0.5 * D22 + D23 + D26 + D30 - D34 - D35 + D36 + D38 + \\
&\quad D39 + D40 - D45 - D46 + D55 \\
C(3,1) &= D4 + D6 + D13 + D14 + D15 + D18 + D24 + D27 + D35 + D36 + D38 + D47 + D52 + D56 - D58 \\
C(3,2) &= 0.5 * D3 + D5 + D7 + 0.5 * D9 + D11 + 0.5 * D17 + D24 + D30 + D35 + D47 + D52 + D74 \\
C(3,3) &= 0.5 * D3 + D5 + D7 + 0.5 * D9 + D11 + 0.5 * D17 + D20 + D21 + D30 + D33 + D36 + D37 + D60 + D70 \\
C(3,4) &= D20 + D21 + D33 + D36 + D37 + D50 + D58 + D60 - D68 \\
C(3,5) &= D13 + D24 + D27 + D35 + D36 + D52 + D56 + D71 \\
C(3,6) &= -D31 + D45 - D54 - D56 + D59 + D61 + D66 + D74 \\
C(3,7) &= -0.5 * D3 - D5 - D7 + 0.5 * D9 - D11 + D12 + 0.5 * D17 + D23 - D30 + D32 - D35 + D39 + D42 + D46 + D68 \\
C(3,8) &= 0.5 * D3 + D5 + D7 + 0.5 * D9 + D11 + D12 + D13 + 0.5 * D17 + D23 + D27 + D30 + D32 - D35 + D36 + D38 + \\
&\quad D39 + D42 - D46 + D61 + D68
\end{aligned}$$

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$$\begin{aligned}
D1 &= (A(1,2) - A(2,1) + A(2,2) + A(2,5) + A(4,2) + A(4,3) - A(4,4) + A(4,5) - A(5,2) - A(5,3) + A(5,4) + A(6,3)) * (B(2,2) - \\
&\quad B(3,2) + B(3,3)) \\
D2 &= (A(2,1) - A(2,2) - A(2,5) + A(3,2) + A(3,3) - A(3,4) + A(3,5) + A(5,2) + A(5,3) - A(5,4) + A(7,2) + A(7,3)) * (-B(2,1) + \\
&\quad B(2,3) + B(3,1)) \\
D3 &= (A(1,2) + A(2,1) + A(5,2) + A(5,3) - A(5,4) + A(7,2) + A(7,3)) * (B(1,1) + B(2,3) - B(3,1)) \\
D4 &= (A(1,2) - A(4,1) + A(5,1) + A(5,2) + A(5,3) - A(5,4) + A(7,2) + A(7,3)) * (B(1,1) + B(2,2) - B(3,1)) \\
D5 &= (A(1,2) + A(5,2) + A(5,3) - A(5,4) + A(7,2) + A(7,3)) * (-B(1,1) + B(3,1)) \\
D6 &= (A(1,2) + A(2,1) - A(3,1) + A(5,2) + A(5,3) - A(5,4) + A(7,2) + A(7,3)) * (B(1,1) + B(2,1) - B(3,1)) \\
D7 &= (A(2,1) - A(2,2) - A(2,5) - A(3,1) + A(3,2) + A(3,5) + A(5,2) + A(5,3) - A(5,4) + A(7,2) + A(7,3)) * (B(2,1) - B(3,1)) \\
D8 &= (A(1,2) + A(1,3) - A(2,1) + A(2,2) + A(2,3) + A(4,2) + A(4,3) - A(4,4) + A(4,5) + A(5,1) - A(5,2) - A(5,3) + A(5,4) + \\
&\quad A(6,3) + A(7,2) + A(7,3)) * (B(3,2) - B(3,3)) \\
D9 &= (-A(2,1) + A(3,1) + A(5,1) + A(7,2) + A(7,3)) * (B(1,2) + B(2,2) - B(3,1)) \\
D10 &= (-A(2,1) + A(5,1) + A(7,2) + A(7,3)) * (-B(1,2) - B(2,2) + B(3,3))
\end{aligned}$$

$D11 = (A(2,1) - A(2,4) - A(2,5) + A(3,2) + A(3,3) - A(3,4) + A(3,5) + A(5,2) + A(5,3) - A(5,4) + A(6,2) - A(6,4) + A(7,2) + A(7,3)) * (B(2,1) - B(2,3))$
 $D12 = (-A(2,1) + A(2,2) + A(2,5) + A(3,1) - A(3,2) - A(3,5) + A(4,1) + A(4,3) - A(4,4) - A(5,2) - A(5,3) + A(5,4) + A(6,3)) * (-B(2,2) + B(3,1))$
 $D13 = (A(2,5) - A(3,5) + A(6,3)) * (-B(2,2) - B(3,1) + B(3,2) + B(5,2))$
 $D14 = (A(1,2) + A(2,4)) * (B(2,1) - B(2,3) + B(3,1) + B(4,1))$
 $D15 = (-A(2,4) + A(3,4) + A(5,1) + A(5,3) + A(7,2) + A(7,3)) * (-B(2,1) + B(2,3) + B(3,1) + B(4,3))$
 $D16 = (A(1,2) - A(2,1) + A(2,2) + A(2,5) + A(4,2) - A(4,4) + A(4,5) + A(5,1) - A(5,2) + A(5,4) + A(7,2)) * (-B(2,2) + B(2,3) + B(3,2) - B(3,3))$
 $D17 = (A(2,5) + A(6,3)) * (-B(2,2) + B(3,2) - B(3,3) + B(5,2))$
 $D18 = (A(2,1) - A(2,2) - A(2,5) + A(5,2) + A(5,3) - A(5,4) + A(7,2) + A(7,3)) * (-B(2,3))$
 $D19 = (A(1,2) + A(4,4)) * (B(2,1) - B(2,2) + B(3,1) + B(4,1))$
 $D20 = (-A(4,4) + A(5,1) + A(5,3) + A(7,2) + A(7,3)) * (-B(2,2) + B(2,3) + B(3,2) + B(4,3))$
 $D21 = (-A(7,2) - A(7,3)) * (B(1,2) + B(2,2))$
 $D22 = (A(4,1) + A(7,2) + A(7,3)) * (B(1,2) + B(2,2) - B(3,2))$
 $D23 = (-A(2,5) + A(3,5)) * (B(2,1) - B(2,2) - B(3,1) + B(3,2) - B(5,1) + B(5,2))$
 $D24 = (-A(1,2)) * (B(2,1) + B(3,1) + B(4,1))$
 $D25 = (-A(2,1) + A(3,1) + A(5,1)) * (B(1,1) - B(1,2) + B(2,1) - B(2,2))$
 $D26 = (A(4,1) + A(4,3) - A(4,4) + A(6,3) + A(7,2) + A(7,3)) * (-B(2,2) + B(3,2))$
 $D27 = (-A(2,5) + A(3,5) + A(6,2) - A(6,4)) * (-B(2,1) + B(5,1))$
 $D28 = (A(1,2) + A(4,2) + A(4,3) - A(4,4) + A(4,5) + A(6,3) + A(7,2) + A(7,3)) * (B(2,2))$
 $D29 = (A(5,1) + A(5,3) - A(5,4) + A(7,2) + A(7,3)) * (-B(2,3) - B(4,3))$
 $D30 = (-A(2,4) + A(5,1) + A(5,3) + A(7,2) + A(7,3)) * (B(3,3) + B(4,3))$
 $D31 = (-A(2,2) - A(2,3) + A(2,4) - A(2,5) + A(3,2) + A(3,3) - A(3,4) + A(3,5) + A(5,2) + A(5,3) - A(5,4) - A(6,3) + A(7,2) + A(7,3)) * (B(3,1))$
 $D32 = (-A(6,3)) * (-B(2,2) + B(3,2) + B(5,2))$
 $D33 = (-A(2,1) + A(5,1) + A(7,2)) * (-B(1,3) - B(2,3) + B(3,3))$
 $D34 = (A(1,2) - A(2,1) + A(2,2) + A(2,5) + A(3,1) - A(3,2) - A(3,5) - A(4,2) + A(4,4) + A(5,1) - A(6,2) + A(6,4)) * (-B(2,1) + B(2,2))$
 $D35 = (A(5,5) - A(6,4)) * (B(2,2) - B(2,3) + B(4,2) - B(4,3) - B(5,1) + B(5,2))$
 $D36 = (A(4,4)) * (-B(2,1) + B(2,2) - B(3,1) + B(3,2) - B(4,1) + B(4,2))$
 $D37 = (-A(2,1) + A(3,1) + A(5,1) + A(7,2)) * (B(1,3) + B(2,1) - B(3,1))$
 $D38 = (A(2,4)) * (-B(2,1) + B(2,3) - B(3,1) + B(3,3) - B(4,1) + B(4,3))$
 $D39 = (A(2,1) - A(5,1)) * (B(1,2) - B(1,3) + B(2,2) - B(2,3))$
 $D40 = (A(1,3) + A(2,3) - A(2,5) - A(3,3) + A(3,5) - A(4,1) - A(4,3) + A(4,4) + A(5,1) - A(6,3)) * (B(3,1) - B(3,2))$
 $D41 = (A(2,1) - A(2,4) + A(3,3) + A(5,3) + A(7,3)) * (B(2,1) - B(2,3) - B(3,1) + B(3,3))$
 $D42 = (A(2,5) - A(6,2) + A(6,4)) * (-B(2,3) + B(5,1))$
 $D43 = (A(2,5)) * (B(2,2) - B(2,3) - B(3,2) + B(3,3) - B(5,2) + B(5,3))$
 $D44 = (A(1,2) + A(2,4) - A(3,4)) * (B(3,1) + B(4,1))$
 $D45 = (-A(4,1) + A(5,1)) * (-B(1,1) + B(1,2) + B(3,1) - B(3,2))$
 $D46 = (A(2,4) - A(3,4) - A(5,4) + A(6,4)) * (-B(2,1) + B(2,3) - B(4,1) + B(4,3))$
 $D47 = (A(5,5)) * (B(2,2) - B(2,3) + B(4,2) - B(4,3) + B(5,2))$
 $D48 = (A(1,1) - A(1,3) + A(1,4) - A(1,5)) * (B(4,2))$
 $D49 = (A(4,4) - A(5,4) + A(5,5)) * (-B(2,2) + B(2,3) - B(4,2) + B(4,3))$
 $D50 = (A(1,3) + A(2,4)) * (B(3,3) + B(4,2))$
 $D51 = (-A(6,1) + A(6,2) - A(6,4) + A(6,5)) * (B(5,1))$
 $D52 = (A(7,1) - A(7,2) + A(7,4) - A(7,5)) * (B(1,3))$
 $D53 = (A(4,5) - A(6,2) + A(6,4)) * (-B(2,2) + B(5,1))$
 $D54 = (-A(1,3) - A(2,4) + A(3,4)) * (B(3,1) + B(4,2))$
 $D55 = (A(1,1) - A(1,5)) * (B(1,2) - B(4,2))$
 $D56 = (A(4,1) + A(7,2)) * (B(1,3) + B(2,2) - B(3,2))$
 $D57 = (A(6,4)) * (B(2,2) + B(4,2) - B(5,1) + B(5,2))$
 $D58 = (A(2,1) + A(2,3) - A(2,4) + A(6,3)) * (B(3,3))$
 $D59 = (A(1,1) - A(1,5) + A(7,4)) * (B(1,3) - B(4,2))$
 $D60 = (-A(2,1) + A(3,1) + A(5,1) + A(6,1) + A(7,1)) * (B(1,1) - B(1,3))$
 $D61 = (-A(1,1) + A(1,5) + A(4,1) + A(7,1)) * (B(1,2) - B(1,3))$
 $D62 = (A(6,1) + A(7,5)) * (B(1,3) + B(5,1))$
 $D63 = (A(1,5) + A(6,1)) * (B(1,2) + B(5,1))$
 $D64 = (A(1,3) + A(4,4)) * (B(3,2) + B(4,2))$

$$\begin{aligned}
D65 &= (A(1,5) + A(4,5) - A(6,4) + A(6,5)) * (-B(5,1) + B(5,2)) \\
D66 &= (A(4,5) + A(6,3)) * (-B(2,2) + B(5,2)) \\
D67 &= (A(2,1)) * (-B(1,1) + B(1,3) + B(3,1) - B(3,3)) \\
D68 &= (A(2,5) - A(5,5) + A(6,5) + A(7,5)) * (-B(5,1) + B(5,3)) \\
D69 &= (-A(6,1)) * (B(1,1) + B(5,1)) \\
D70 &= (A(1,4) + A(2,4) - A(3,4)) * (B(4,1) - B(4,2)) \\
D71 &= (A(1,4) + A(2,4) - A(7,4)) * (-B(4,2) + B(4,3)) \\
D72 &= (A(6,4)) * (B(2,1) + B(4,1)) \\
D73 &= (-A(1,5)) * (B(1,2) + B(5,2)) \\
D74 &= (A(7,5)) * (B(1,3) + B(5,3)) \\
D75 &= (A(7,4)) * (-B(1,3) + B(4,3)) \\
D76 &= (A(2,5) - A(4,5)) * (B(2,3) - B(5,3)) \\
D77 &= (A(1,1) + A(6,1)) * (B(1,1) - B(1,2)) \\
D78 &= (A(7,4)) * (-B(1,3) + B(4,1)) \\
D79 &= (A(5,5)) * (B(5,3)) \\
D80 &= (A(3,5)) * (-B(2,3) + B(5,3)) \\
D81 &= (A(7,5)) * (B(1,3) + B(5,2)) \\
D82 &= (A(1,5)) * (B(1,3) + B(5,3))
\end{aligned}$$

$$\begin{aligned}
C(1,1) &= -D24 - D44 + D48 - D54 + D55 + D63 + D69 + D70 + D77 \\
C(1,2) &= -D19 - D24 - D36 + D48 + D55 + D64 - D73 \\
C(1,3) &= -D14 - D24 - D38 + D48 + D50 + D59 + D71 + D75 + D82 \\
C(2,1) &= D2 + D3 + D5 + D11 + D13 + D14 + D18 + D23 + D24 + D27 - D31 + D32 + D42 \\
C(2,2) &= D1 + D4 + D5 + D8 + D10 + D17 + D19 + D22 + D24 - D28 + D32 + D36 + D45 + D50 + D58 - D64 \\
C(2,3) &= D3 + D5 + D14 + D17 + D18 + D24 + D32 + D38 + D43 + D58 + D67 \\
C(3,1) &= D2 + D3 - D6 + D7 + D11 + D14 + D18 + D27 + D42 - D44 \\
C(3,2) &= D1 + D8 + D9 + D10 + D12 - D13 + D17 - D26 - D28 + D31 + D40 + D50 + D54 + D58 \\
C(3,3) &= D2 + D7 + D15 + D18 - D30 + D33 + D37 + D41 + D80 \\
C(4,1) &= -D4 + D6 - D7 + D9 + D12 + D13 + D19 + D21 + D23 + D24 + D25 + D27 + D28 + D32 + D34 + D53 \\
C(4,2) &= D19 + D21 + D22 + D24 + D26 + D28 + D32 + D36 + D66 \\
C(4,3) &= D1 - D10 + D14 + D16 + D17 - D18 - D20 + D21 + D24 + D26 + D30 + D32 + D33 + D38 + D39 + D43 + D56 + D76 \\
C(5,1) &= D5 + D6 + D9 + D15 + D21 + D24 + D25 + D29 - D35 + D44 + D46 + D47 - D57 + D72 \\
C(5,2) &= D4 + D5 + D19 + D20 + D21 + D22 + D24 + D29 + D36 + D45 + D47 + D49 \\
C(5,3) &= D3 + D5 - D10 + D14 + D21 + D24 + D29 + D30 + D38 + D39 + D67 + D79 \\
C(6,1) &= -D13 - D23 - D27 - D32 + D51 - D69 + D72 \\
C(6,2) &= -D32 + D51 + D53 + D57 + D63 + D65 - D66 + D73 \\
C(6,3) &= -D17 - D32 + D35 + D42 - D43 - D47 + D51 + D57 + D62 + D68 - D74 + D79 \\
C(7,1) &= -D9 - D21 - D25 + D37 + D52 + D60 + D62 + D69 + D78 \\
C(7,2) &= -D21 - D22 + D52 + D55 + D56 - D59 + D61 + D81 \\
C(7,3) &= D10 - D21 - D33 - D39 + D52 + D74 + D75
\end{aligned}$$

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